



1.2 RESISTOR DISSIPATION

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If the flow of current through a resistor increases, it heats up, and if the temperature exceeds a certain critical value, it can be damaged. The wattage rating of a resistor is the power it can dissipate over a long period of time.

Wattage rating is not identified on small resistors. The following diagrams show the size and wattage rating:

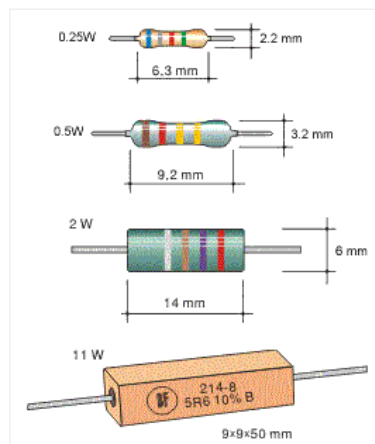


Fig. 1.3: Resistor dimensions

Most commonly used resistors in electronic circuits have a wattage rating of 1/2W or 1/4W. There are smaller resistors (1/8W and 1/16W) and higher (1W, 2W, 5W, etc).

In place of a single resistor with specified dissipation, another one with the same resistance and higher rating may be used, but its larger dimensions increase the space taken on a printed circuit board as well as the added cost.

Power (in watts) can be calculated according to one of the following formulae, where U is the symbol for Voltage across the resistor (and is in Volts), I is the symbol for Current in Amps and R is the resistance in ohms:

$$\begin{aligned} P &= U \cdot I \\ P &= R \cdot I^2 \\ P &= \frac{U^2}{R} \end{aligned}$$

For example, if the voltage across an 820Ω resistor is 12V, the wattage dissipated by the resistors is:

$$P = \frac{U^2}{R} = \frac{12^2}{820} = 0,176 \text{ W} = 176 \text{ mW}$$

A 1/4W resistor can be used.

In many cases, it is not easy to determine the current or voltage across a resistor. In this case the wattage dissipated by the resistor is determined for the “worst” case. We should assume the highest possible voltage

across a resistor, i.e. the full voltage of the power supply (battery, etc).

If we mark this voltage as V_B , the highest dissipation is:

$$P = \frac{U_B^2}{R}$$

For example, if $V_B=9V$, the dissipation of a 220W resistor is:

$$P = \frac{9^2}{220} = 368 \text{ mW},$$

A 0.5W or higher wattage resistor should be used



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